

Mahmoud Gamal

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Professional Summary

Final-year Computer Science student focused on Data Engineering, with hands-on experience building and automating data pipelines and processing real-world datasets through academic projects and technical training. Strong foundation in data engineering concepts and practical experience developing reliable data solutions. Eager to contribute to building high-quality data workflows that support analytics and business needs.

Training

Microsoft Data Engineer Internship – Digital Egypt Pioneers Initiative (DEPI) Jul 2026 – Present

- Covered database development, data warehousing, Python programming, data analysis, Azure data fundamentals, and data engineering concepts through hands-on coursework and a capstone project.

Big Data Summer Internship – National Telecommunication Institute (NTI) Aug 2026 – Sep 2026

- Worked on distributed data processing and streaming workflows through practical exercises and a hands-on big data project.

Projects

Stock Market Data Pipeline (Python, Airflow, PostgreSQL, Docker)

- Built a daily **incremental ETL pipeline** for **8 public companies** using Python and Yahoo Finance, with PostgreSQL as the analytical database and a Star Schema for structured reporting.
- Automated workflow execution with **Apache Airflow**, using incremental upserts, **3 retries**, and a 5-minute retry delay to support reliable scheduled runs and safe reruns.

Retail Store Data Pipeline (Python, Pandas, SQL Server)

- Built an end-to-end **ETL pipeline** to clean, transform, and load **9 raw CSV files** containing **10,000+ records** into SQL Server.
- Designed a **3NF SQL Server database with 9 tables** and executed **11+ analytical queries** to analyze sales, products, and customers, revealing **6.80M in revenue from Baldwin Bikes**.

Technical Skills

Programming: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL

Data Engineering & Orchestration: ETL/ELT, Data Pipelines, Data Warehousing, Dimensional Modeling, Apache Airflow

Big Data & Streaming: Hadoop, Apache Spark, Apache Kafka

Cloud & Data Platforms: Microsoft Azure, Microsoft Fabric (OneLake, Lakehouse)

Databases & BI: PostgreSQL, Excel, Power BI

DevOps & Tools: Docker, Git, Linux (Bash)

Certifications

- **Microsoft Certified: Azure Data Fundamentals (DP-900)**

Leadership & Activities

Associate Student Ambassador – Microsoft Jul 2026 – Present

- Part of Microsoft's global student ambassador program, learning Microsoft technologies, engaging with the developer community, and contributing through technical activities.

Data Engineer Member – Microsoft Student Club, KFS Oct 2025 – Aug 2026

- Learned and applied **Data Engineering** through practical tasks, hands-on projects, and technical reviews; selected as **Best Member**, and my project was awarded **Best Project**.

Education

B.S. in Computer Science – Mansoura University Oct 2023 – May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Database Systems, Object-Oriented Programming.

Languages

Arabic: Native **English:** Professional Working Proficiency